

Multiple Choice

1. **Answer:** B

Options: A) Larger if the error rate is larger.; B) Larger if the error rate is smaller.; C) Smaller if the error rate is smaller.; D) It does not matter.

Note: Correct option: B - Larger if the error rate is smaller.

2. **Answer:** B

Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

Note: Correct option: B - Improve the optimisation algorithm being used for error minimisation.

3. **Answer:** A

Options: A) random crop and horizontal flip; B) random crop and vertical flip; C) posterization; D) dithering

Note: Correct option: A - random crop and horizontal flip

4. **Answer:** A

Options: A) The polynomial degree; B) Whether we learn the weights by matrix inversion or gradient descent; C) The assumed variance of the Gaussian noise; D) The use of a constant-term unit input

Note: Correct option: A - The polynomial degree

5. **Answer:** D

Options: A) It can not be applied to non-differentiable functions.; B) It can not be applied to non-continuous functions.; C) It is hard to implement.; D) It runs reasonably slow for multiple linear regression.

Note: Correct option: D - It runs reasonably slow for multiple linear regression.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$ '.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def first_repeated_word(str 1): | return word; | return 'None'`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def count_vowels(test_str): | return (res)`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key